

ANKAN DAS

Bengaluru, India (Open to relocate to Germany)

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Summary

Backend and distributed systems engineer with 5+ years of experience delivering high-scale, low-latency platforms across AI and fintech domains. Specialized in cloud-native architectures, event-driven systems, and microservices with strong ownership across system design, production reliability, and cost optimization. Proven track record of scaling mission-critical services, reducing latency, and mentoring engineers in fast-paced environments.

Core Skills

- **Languages & Backend:** Python, Java, JavaScript, SQL, Django, Flask, FastAPI
- **Cloud & DevOps:** AWS (Lambda, SQS, S3, EC2, DynamoDB), Docker, Kubernetes, Terraform
- **Architecture:** Distributed Systems, Microservices, Event-Driven Design, API Design
- **Scalability:** Load Balancing, Caching, CAP Theorem, Async Processing
- **Security:** Zero-Trust Architecture, mTLS, HMAC Auth, Model Encryption
- **Observability:** Prometheus, Grafana, Logging & Monitoring

Experience

Murf AI

November 2022 - Present

Software Development Engineer II

Bengaluru, India

- Architect and own distributed TTS platforms processing 50M+ daily requests across 30+ languages with sub-150ms latency.
- Optimize microservices by consolidating G2P and TTS workflows within the same GPU instance, reducing costs by 35%
- Design zero-trust security for on-premises deployments using ONNX model encryption, cythonization, mTLS, and HMAC authentication.
- Build containerized local license agent enabling secure usage metering and credit validation for air-gapped customers.
- Implement AWS-based CI/CD with automated testing, security scans, and blue-green deployments, cutting release cycles by 75%.
- Mentor and lead junior engineers through reviews, architecture guidance, and production best practices.

Mewt Accounts Pvt Ltd (SuperPe)

September 2021 - May 2022

Backend Developer

Bengaluru, India

- Built backend services for bank account verification and transaction tracking in regulated fintech workflows.
- Implemented asynchronous pipelines using Kafka, Celery, and AWS Lambda to process 500+ events/second reliably.
- Optimized APIs and databases to support 10M+ daily records with low-latency responses.

Projects

EventHorizon ([GitHub Link](#)): Event-Driven E-Commerce Platform

- Designed event-driven backend architecture to support high-throughput e-commerce workflows.
- Implemented Kafka-based pipelines with FastAPI, Postgres, and Redis for fault tolerance and scalability.
- Improved system resilience and throughput for multi-service transactional processing.

FloodGate ([GitHub Link](#)): Multi-Tenant Distributed Rate Limiting Gateway

- Built a multi-tenant rate-limiting gateway to protect backend services under extreme traffic spikes.
- Implemented Redis-backed coordination with token bucket algorithms for horizontal scalability.
- Sustained 50K+ RPS with <10ms p99 latency and <0.1% error rate under load testing.

AegisFL ([GitHub Link](#)): Privacy-Preserving Federated Learning Platform

- Built an end-to-end federated learning platform using TensorFlow Federated across 1,000+ simulated clients on CIFAR-10.
- Designed federated training with secure aggregation and differential privacy considerations to align with GDPR and HIPAA data-protection principles, achieving 96% accuracy.
- Deployed on AWS EKS with mTLS-secured gRPC and Prometheus/Grafana, reducing convergence time by 30% from baseline.

StreamPulse ([GitHub Link](#)) - Real-Time Anomaly Detection System

- Developed real-time streaming platform for anomaly detection across high-volume event datasets.
- Integrated FastAPI inference pipelines delivering less than 200 milliseconds latency at peak load.
- Implemented monitoring with Prometheus and Grafana for live system observability.

Education

Indian Institute of Technology Kanpur

July 2017 - May 2021

B.Tech in Electrical Engineering

Kanpur, India

Languages

- English: Full Professional Proficiency (C2)
- German: Intermediate Working Proficiency (B1)